# Demonstration of Proposed Features

| | |
|---|---|
| **Date** | 28 June 2026 |
| **Team ID** | |
| **Project Name** | Rising Waters: A Machine Learning Approach to Flood Prediction |
| **Maximum Marks** | 1 Mark |

### Demonstration of Proposed Features:

This document summarizes the features proposed during the planning phase of the **Rising Waters** project and records their implementation status. It demonstrates that the proposed machine learning solution for flood prediction has been successfully developed, tested, and demonstrated using a Flask-based web application.

| S.No | Feature Name | Description | Status (Implemented / Partial / Pending) | Demonstrated (Yes / No) | Remarks |
|---|---|---|---|---|---|
| 1 | Dataset Loading | Load historical flood prediction dataset using Pandas | Implemented | Yes | Dataset successfully loaded and verified |
| 2 | Exploratory Data Analysis (EDA) | Perform descriptive statistics, distribution plots, box plots, and correlation heatmap | Implemented | Yes | Visualizations generated successfully |
| 3 | Data Preprocessing | Handle missing values, treat outliers, perform feature scaling, and split dataset | Implemented | Yes | Dataset prepared succes |
| 4 | Machine Learning Model Training | Train Decision Tree, Random Forest, KNN, and XGBoost models | Implemented | Yes | All four models trained and evaluated |
| 5 | Model Evaluation & Best Model Selection | Compare models using Accuracy Score, Confusion Matrix, and Classification Report | Implemented | Yes | Best-performing model selected and saved |
| 6 | Flask Web Application | Develop a web interface to predict flood occurrence using the trained model | Implemented | Yes | Prediction system demonstrated successfully |

**Feature Implementation Summary:**

| | |
|---|---|
| **Total Features Proposed** | 6 |
| Total Features Implemented | 6 |
| Total Features Demonstrated | 6 |
| Overall Implementation Rate (%) | 100% |